# Vaibhav Saran

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## Summary

AI Engineer with hands-on experience building and deploying production agentic AI systems, from usage-analytics and diagnostic skills on an enterprise multi-agent platform to autonomous race-intelligence pipelines and interpretable clinical RL frameworks. Ships end-to-end ML systems across agentic AI, data engineering and MLOps, with published research in offline RL & sustainable ML.

## Education

**University at Buffalo, The State University of New York (SUNY)** Dec 2025
*M.S., Artificial Intelligence* *Buffalo, NY*
- GPA: 3.7/4.0

**JIET Group of Institutions** Jun 2022
*B.Tech, Computer Science Engineering* *Jodhpur, RJ*
- GPA: 3.64/4.0

## Experience

**Trek Health** Aug 2026 – Present
*AI/ML Engineering Intern* *San Mateo, CA*
- Cut error-resolution time by 70% for sales teams and customers by designing and deploying a diagnostic skill that traces failed agent runs to a priority-ranked root cause using an evidence-bundle query and mandatory sub-agent drilldown.
- Improved response time by 87% and cut model calls by 38% by migrating the agent platform's LangGraph orchestration architecture to a REST API architecture which led to reduction in infrastructure costs by 42%.

**Green AI Labs, University at Buffalo** Jan 2025 – Aug 2026
*Research Intern* *Buffalo, NY*
- Improved patient treatment accuracy to 83%, a 23-point gain over the BCQ baseline, by building an LLM-grounded decision-support system that generates natural-language treatment rationale from RAG-retrieved clinical knowledge, layered on an ensembled offline RL, XGBoost, and TabNet policy trained over 874K ICU records.
- Cut patient admission-time triage delay by classifying incoming ICU cases into 3 validated mortality-risk tiers within minutes of admission, via a clustering-based risk stratification module (HDBSCAN, UMAP) built in Python over 27,799 patient stays, flagging high-risk cases for physician priority intervention ahead of clinician availability.

**Innomatics Research Labs** Jan 2023 – Jun 2024
*Data Scientist* *Hyderabad, TG*
- Validated feasibility of consolidating multi-retailer pricing at scale by building a PoC price aggregator for Amazon, BestBuy, and Flipkart. Engineered a Python scraping pipeline (BeautifulSoup, Selenium) collecting over 5000 listings via automated weekly refresh, and stress-tested it under a simulated production-scale emulation.
- Normalized inconsistent multi-source data into 12 structured attributes by building source-specific cleaning modules and a distributed PySpark ETL pipeline into PostgreSQL, surfacing real-time CRUD and cross-brand price analytics through a unified interface.

## Projects

**StockPilot AI** Feb 2026 – Aug 2026
- Automated multi-source stock research into one query, cutting response time under 10 seconds, by building a multi-agent supervisor system with hybrid-search routing, hardened by an 11.4% data-quality filter and a prompt-injection guardrail.
- Extended stock analysis with 7-day Prophet-based price forecasting while archiving 3,800+ cold records to Amazon S3 with automatic DuckDB fallback, orchestrated via Airflow across PostgreSQL, MongoDB, and ChromaDB.

**F1 Primus AI** Jan 2026 – Feb 2026
- Cut pre-race analysis time to under 2 minutes by building a Python LangGraph ReAct agent that autonomously aggregates and reasons over weather, practice, and news data across 24 rounds, combined with a Gradient Boosting lap-time predictor.
- Built a reproducible MLOps pipeline using MLflow for offline experiment tracking and Prefect for orchestration with automated retries, containerized with Docker and deployed to GCP Cloud Run via GitHub Actions CI/CD.

## Publications

Kumar P., Saran V., Patel D., Kulkarni N., Vereshchaka A. Attention-Based Offline RL and Clustering for Interpretable Sepsis. IEEE ICMLA 2025. 83% accuracy on MIMIC-III & eICU via ensemble offline RL with custom attention mechanism.

Saran V., Imran A. Green CRISP-ML(Q): A Framework for Sustainable Machine Learning. Poster. NY-BEST Annual Energy Storage Conference, 2025, Ithaca, NY.

## Skills & Tools

**Programming Languages**: Python, C++, C
**AI/ML**: LangChain, LangGraph, RAG, LangSmith, Multi-Agent Systems, Claude API, OpenAI API, Gemini, MCP, PyTorch, scikit-learn, Hugging Face, Reinforcement Learning (RL), VAE, Diffusion Models, XGBoost, HDBSCAN, UMAP, NLTK, SpaCy, Speech/Audio Processing
**Data Engineering**: Apache Airflow, Prefect, PySpark, Hadoop, FastAPI, Snowflake, Databricks, Microsoft Fabric
**Databases**: PostgreSQL, MongoDB, ChromaDB, FAISS, MySQL, SQLite
**AI Inference and Serving**: AWS (EC2, S3), GCP, Groq, Cerebras, vLLM, TensorRT
**Tools & DevOps**: MLflow, Docker, GitHub Actions CI/CD, Git, Selenium, Nginx